Evaluating AI:

Innovation, Ethics and the Limits of Governance

What AI Is

Machine Learning

Systems trained on data to identify patterns and make predictions — without being explicitly programmed for each decision.

Neural Networks

Layers of interconnected nodes that process information similarly to the human brain, enabling image recognition, language processing, and complex classification.

Large Language Models

AI trained on vast text to generate, translate, and reason about language. Foundation of tools such as ChatGPT and enterprise AI assistants.

Common thread: AI makes or informs consequential decisions at a speed and scale that no human review process can match.

REAL-WORLD APPLICATIONS

AI in Enterprise

Fraud Detection

Real-time transaction scoring in banking and payments. AI flags anomalies in milliseconds — faster than any analyst.

Medical Diagnosis

AI-assisted radiology, pathology, and early disease detection. In trials, outperforms specialists on specific classification tasks.

Identity Verification

Biometric matching and document authentication. Reduces manual review time while scaling to millions of checks daily.

Hiring Algorithms

CV screening and candidate ranking at scale. Used by majority of Fortune 500 firms — and the source of the most significant failures.

Sources: Chang (2022); Manure, Bengani and Saravanan (2023)

ETHICAL CONCERNS

Three Persistent Problems

01

Bias

Models trained on historical data encode historical inequalities. The output reflects — and amplifies — what was already unfair.

02

Opacity

When AI denies a loan, a job, or a visa, the applicant cannot ask why. The model's reasoning is not accessible to those it affects.

03

Accountability gap

When the model is wrong, responsibility is diffuse. Developer, deployer, or training data — the question of who is liable remains unresolved in most jurisdictions.

Sources: Manure, Bengani and Saravanan (2023); Walton (2024); Bashynska and Prokopenko (2024)

AI as an Attack Surface

Adversarial attacks

Inputs deliberately crafted to fool the model. A stop sign with small stickers causes an autonomous vehicle to classify it as a speed limit sign. The model is confident and wrong.

Data poisoning

Corrupting training data to cause systematic misclassification. An attacker who can influence what an AI learns can control what it decides — potentially, at scale.

Model theft

Reverse-engineering proprietary models through repeated querying. The same openness that makes AI useful for users makes it exploitable by adversaries.

Sources: Bashynska and Prokopenko (2024); Nielson (2023)

The Governance Response

GDPR 2018	**Data protection — EU** Article 22: right not to be subject to solely automated decisions with significant effects. Requires transparency, human review on request.
EU AI Act In force Aug 2024	**AI regulation — EU** Risk-based classification: prohibited, high-risk, limited, minimal. High-risk includes biometric ID, credit scoring, employment decisions. Fully applicable 2026.
NIST AI RMF 2023	**Voluntary framework — US** Govern / Map / Measure / Manage. Provides enterprise-level structure for identifying, assessing, and mitigating AI risk across the development lifecycle.

Sources: EU AI Act, Regulation (EU) 2024/1689; Manure, Bengani and Saravanan (2023); Bashynska and Prokopenko (2024)

CASE STUDIES

When AI Gets It Wrong

Amazon — Recruitment algorithm (2018)

System scrapped

Trained on a decade of hiring decisions, the model downgraded CVs containing the word "women's" and penalised graduates of all-female colleges. Neither bias nor any correction was visible to applicants. Amazon scrapped the system.

Ofqual (UK) — A-level exam algorithm (2020)

Algorithm reversed

AI awarded grades based on schools historical performance. 40% of predicted A-level grades were downgraded. Students from disadvantaged schools were systematically penalised. The algorithm was reversed within days under public and political pressure.

Sources: Manure, Bengani and Saravanan (2023); Fu et al. (2025)

Balancing Innovation and Responsibility

1

Classify before deploying

Apply the EU AI Act risk tiers. Know whether a system is high-risk before it reaches production.

2

Human oversight on consequential decisions

Automated decisions affecting access to credit, employment, or services require a human review mechanism.

3

Document the model

Training data sources, known limitations, performance by demographic group. If it cannot be documented, it is not ready to deploy.

4

Regular bias audits

Models drift as data changes. A system that was fair at deployment may not be fair six months later.

5

Treat GDPR and EU AI Act as a floor

Compliance is the minimum. Organisations that embed governance into development — not bolt it on afterwards — build more durable systems.

Sources: Manure, Bengani and Saravanan (2023); Bashynska and Prokopenko (2024); EU AI Act (2024)

Lethal Autonomous Weapons Systems

— Definition (DoD Directive 3000.09, 2023): "Weapon systems that, once activated, can select and engage targets without further intervention by a human operator."

— AI-guided autonomous weapons are already deployed in active conflicts. Ukraine has been described as the first combat testing ground for LAWS.

— 2 December 2024: UN General Assembly passed a resolution on LAWS — 166 votes in favour, 3 opposed: Russia, North Korea, and Belarus.

— "Human-in-the-loop" is becoming a phrase used to reassure the public. In practice, it risks meaning a human presses a go button with minimal insight into what the system is doing or why.

— The EU AI Act does not apply to military systems. GDPR does not apply to national security. The NIST framework is voluntary.

Sources: Russell (2023) Nature; Longpre et al. (2022) MIT Science Policy Review; DoD Directive 3000.09 (2023); UN General Assembly Resolution (Dec 2024)

The Gap

We have built frameworks — GDPR, the EU AI Act, the NIST RMF — serious, carefully constructed attempts to make AI accountable. They cover enterprise AI: hiring algorithms, credit scores, medical diagnosis.

They do not cover the domain where AI is making the most irreversible decisions of all.

The opacity is the same. The accountability gap is the same. The potential for error is the same. The consequences are not.

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